

Multi-Modal, Training-Free Crack Extraction via Generalized Frangi Graph

Louis HAUSEUX*, Raphaël ANTOINE[°], Philippe FOUCHER[°], Pierre CHARBONNIER[°], and Josiane ZERUBIA*

^{*}Inria, Université Côte d’Azur, Ayana team, France

[°]Cerema, ENDSUM research team, France

Corresponding author: louis.hauseux@inria.fr

Abstract—Automatic crack extraction is essential for both infrastructure inspection and geosciences, but annotated crack data remain scarce and domain shifts are frequent. We address the problem of extracting a crack network from one or several scalar modalities, without training.

To do so, we generalize the classical FRANGI vesselness filter from a pixel-wise operator to a pairwise spatial graph. While the standard filter is limited to independent pixel-wise geometry, our formulation enriches this description by explicitly encoding the orientation alignment between neighboring pixels, alongside elongation and contrast. The final crack network is then **resolved via a graph extraction—extracted through a graph pipeline** combining minimum spanning tree reduction and similarity-weighted **betweenness-tree** centrality.

While supervised models perform best on clean, in-distribution data, our geometry-driven framework **exhibits strong robustness to noise. Its ability to generalize without training across large domain shifts is validated on real-world datasets (from civil infrastructures to geological images)degrades more slowly than CrackSegDiff under the tested synthetic perturbations.**

Index Terms—Crack detection, multi-modal fusion, FRANGI filter, graph, minimum spanning tree, **betweenness-tree** centrality

I. INTRODUCTION

Reliable crack mapping is needed for civil infrastructure inspection and for monitoring geological hazards such as landslides, faults and subsidence [1]. In practice, many crack maps are still produced or corrected manually, which is slow, subjective and difficult to scale. Automatic extraction is difficult because cracks are thin, discontinuous, and are often low-contrast structures whose appearance varies with the sensor, material, illumination, surface roughness and acquisition geometry.

Deep learning methods have become the dominant approach for crack segmentation. They perform well when **train-training** and test images follow the same distribution, but they require annotated data and may fail under sensor or domain shifts [2]–[4]. This limitation is especially relevant for multi-modal crack analysis, where public benchmarks with **well-aligned, clean, clean, well-aligned** visible, depth or thermal modalities are still rare.

This paper takes a complementary direction: our framework provides an interpretable, **domain-agnostic-training-free** extraction pipeline. It addresses multi-modal scenarios for which annotations are not available. In **such contextthis setting**, creating ground truth is very expensive. Our method can di-

rectly assist domain experts (*e.g.*, geophysicists) and accelerate dataset annotation.

Our main contributions are as follows: (1) We generalize the FRANGI filter [5] from a pixel-wise vesselness response to a pairwise graph similarity using second-order shape, contrast and alignment terms, **together with normalized multi-modal Hessian fusion**; (2) We evaluate our complete algorithm on the FIND benchmark [6] and on two geological image sets, including comparisons with supervised models on clean and noisy data. **The contribution lies in coupling these established components through the pairwise FRANGI graph.**

II. RELATED WORK

A. Classical Curvilinear Filters

Multi-scale Hessian filters are a standard tool for enhancing curvilinear structures. The classical FRANGI vesselness filter [5] estimates whether a pixel belongs to a tubular structure from the eigenvalues of a smoothed Hessian matrix, following related work such as **SatoSATO**’s line filter [7]. These methods are fast and interpretable, but they are local: **They-they** evaluate pixels independently and can fragment under noise or textured backgrounds. Our method keeps the Hessian geometry but moves the decision from isolated pixels to edges of a spatial graph. **Our preliminary study [8] considered two single-modal geological images. Here we add normalized multi-modal fusion, evaluate clean and noisy FIND data, and include a second geological setting.**

B. Deep Crack Segmentation

Deep learning architectures such as U-Net [9], DeepCrack [10], CrackFormer [11] and recent diffusion models such as CrackSegDiff [12] have achieved strong crack segmentation results. Their main practical weakness is **the-dependency-their dependence** on annotations. A recent survey by **Zhang-ZHANG et al.** [4] also emphasizes poor cross-domain robustness when sensors, materials or acquisition conditions change. This motivates training-free alternatives and hybrid strategies that can operate when annotations are not available.

C. Multi-Modal Fusion

In learned systems, multi-modal fusion is usually implemented by channel concatenation, intermediate feature fusion or late prediction fusion. These choices depend on the network architecture and on the modalities available at training time.

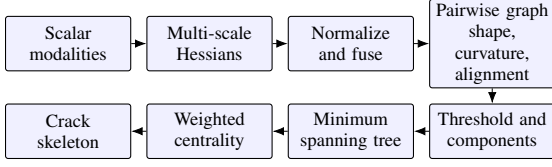


Fig. 1: **Method overview.** Local Hessian cues are fused across modalities and lifted to pairwise graph similarities; graph reduction and weighted centrality return the crack skeleton.

Here, fusion is performed at the operator level: Each modality contributes a normalized Hessian, and the fused Hessian is processed by the same graph pipeline regardless of the number of modalities.

III. METHODOLOGY

A. Problem Formulation and Skeleton Output

Let $\Omega \subset \mathbb{Z}^2$ be the image grid and let

$$\mathcal{I} = \{I^{(m)} : \Omega \rightarrow \mathbb{R}\}_{m \in \mathcal{M}} \quad (1)$$

be one or several scalar modalities, such as visible intensity, depth/range or thermal infrared images. The output is a set of pixels $\hat{\Gamma} \subset \Omega$ representing the crack skeleton. We use “skeleton” in the standard image-analysis sense: A one-pixel-wide centerline representation that preserves the main connectivity and branches of a thin object; thick masks used for evaluation are skeletonized with the implementation reported in [13], [14].

The method assumes that the crack of interest is visible as a dark curvilinear ridge. If a sensor encodes cracks as bright ridges, the modality can be inverted before processing. Figure 1 summarizes the processing chain.

B. Multi-Scale Hessian Features

For a modality $I^{(m)}$ and a Gaussian scale σ , the smoothed Hessian at $x \in \Omega$ is

$$\mathcal{H}_\sigma^{(m)}(x) = \begin{pmatrix} \partial_{xx}(G_\sigma * I^{(m)})(x) & \partial_{xy}(G_\sigma * I^{(m)})(x) \\ \partial_{xy}(G_\sigma * I^{(m)})(x) & \partial_{yy}(G_\sigma * I^{(m)})(x) \end{pmatrix}. \quad (2)$$

Equivalently, the derivatives are computed by convolution with derivatives of the Gaussian kernel [5], [15],

$$\partial_{ab}(G_\sigma * I^{(m)}) = (\partial_{ab}G_\sigma) * I^{(m)}, \quad a, b \in \{x, y\}. \quad (3)$$

Since the Hessian matrix in (2) is real and symmetric, we compute its eigenvalues $\lambda_1^\sigma(x), \lambda_2^\sigma(x)$, ordered as $|\lambda_1^\sigma(x)| \leq |\lambda_2^\sigma(x)|$. The eigenvector associated with $\lambda_1^\sigma(x)$ gives the local longitudinal direction of the ridge. For an elongated ridge, transverse curvature dominates longitudinal curvature, so $R_B = |\lambda_1^\sigma|/|\lambda_2^\sigma|$ is near zero; for a blob-like structure, the two magnitudes are comparable and R_B approaches one. Classical FRANGI vesselness combines this geometry with curvature magnitude and sign at each pixel. Our graph retains these local cues while requiring compatible evidence and longitudinal directions at neighboring pixels.

For multi-modal inputs, each Hessian is normalized by the largest spectral norm [16] in the image,

$$\hat{\mathcal{H}}_\sigma^{(m)}(x) = \frac{\mathcal{H}_\sigma^{(m)}(x)}{\max_{y \in \Omega} \|\mathcal{H}_\sigma^{(m)}(y)\|_2}, \quad (4)$$

and the fused Hessian is

$$\mathcal{H}_\sigma^{\text{fused}}(x) = \sum_{m \in \mathcal{M}} w_m \hat{\mathcal{H}}_\sigma^{(m)}(x), \quad w_m \geq 0, \quad \sum_m w_m = 1. \quad (5)$$

All eigenvalues and orientations used below are computed from (5).

~~The geometric assumption behind this operator-level fusion is that the structures of interest (i.e., the cracks) share an identical geometry across all modalities. Conversely, we assume that the crack geometry is shared across modalities, whereas sensor-specific noise and domain artifacts behave as uncorrelated random variables from one modality to another. Consequently, linearly aggregating artifacts are largely uncorrelated. Under this assumption, averaging the normalized Hessians reinforces the consistent structural signal while uncorrelated cross-modal noise is averaged out consistent structures and reduces modality-specific noise.~~

C. Generalized FRANGI Similarity Graph

We build a sparse spatial graph $G = (V, E)$ on candidate pixels $V \subseteq \Omega$. ~~Two pixels~~ We denote the distance between two candidate pixels by

$$\rho_{ij} = \|x_i - x_j\|_2. \quad (6)$$

Pixels $x_i, x_j \in V$ are connected when

$$0 < \|x_i - x_j\| \leq R.$$

$$0 < \rho_{ij} \leq R. \quad (7)$$

The radius R allows short interruptions caused by noise or local loss of contrast to be bridged, while ~~the distance factor introduced below~~ ρ_{ij} discourages long accidental jumps.

For a scale $\sigma \in \Sigma$, an edge (i, j) is assigned a similarity $S_{ij}^\sigma \in [0, 1]$ as a product of three soft constraints.

Elongation. ~~With the FRANGI ratio~~ $R_B(x) = |\lambda_1^\sigma(x)|/|\lambda_2^\sigma(x)|$, ~~both~~ Both endpoints must have ~~line-like second-order geometry~~ a small FRANGI ratio:

$$S_{\text{shape}}^\sigma(i, j) = \exp \left[-\frac{1}{2} \left(\frac{R_B(x_i) + R_B(x_j)}{s_s} \right)^2 \right]. \quad (8)$$

Positive curvature. For dark ridges, the useful transverse curvature is $C_\sigma(x) = \lambda_2^\sigma(x) \mathbf{1}_{\lambda_2^\sigma(x) > 0}$. The intensity term is

$$S_{\text{int}}^\sigma(i, j) = 1 - \exp \left[-\frac{1}{2} \left(\frac{\sqrt{C_\sigma(x_i)C_\sigma(x_j)}}{s_i} \right)^2 \right]. \quad (9)$$

The geometric mean makes the edge weak when either endpoint has weak curvature.

Orientation alignment. If $\theta^\sigma(x)$ is the angle of the longitudinal eigenvector, direction compatibility is modeled by

$$S_{\text{align}}^\sigma(i, j) = \exp \left[-\frac{1}{2} \left(\frac{\sin(\theta^\sigma(x_i) - \theta^\sigma(x_j))}{s_a} \right)^2 \right]. \quad (10)$$

The sine accounts for the fact that crack directions are unoriented: θ and $\theta + \pi$ describe the same local axis.

~~The final multi-scale similarity and the associated dissimilarities are~~

$$S_{ij} = \max_{\sigma \in \Sigma} S_{\text{shape}}^\sigma(i, j) S_{\text{int}}^\sigma(i, j) S_{\text{align}}^\sigma(i, j),$$

We first take the strongest similarity across scales,

$$\bar{S}_{ij} = \max_{\sigma \in \Sigma} S_{\text{shape}}^\sigma(i, j) S_{\text{int}}^\sigma(i, j) S_{\text{align}}^\sigma(i, j), \quad (11)$$

$$d_{ij} = (1 - S_{ij}) \|x_i - x_j\|.$$

then include spatial distance in the graph similarity and dissimilarity:

$$S_{ij} = \max\{0, 1 - (1 - \bar{S}_{ij})\rho_{ij}\}, \quad d_{ij} = 1 - S_{ij}. \quad (12)$$

Thus the graph edge is strong only when both pixels are elongated, have the correct curvature sign and share a compatible orientation. The ~~distance-weighted dissimilarity factor~~ ρ_{ij} in (12) regularizes the later tree extraction.

~~The parameters are fixed to~~ For FIND and the Palais des Papes, we use $s_s = 2$, $s_i = 1/4$ and $s_a = 1/8$ ~~in the FIND experiments, with~~, $\Sigma = \{1, 3, 5, 7\}$ and $R = 3$. For the single-modal Vaches Noires images, the ~~scale set is slightly enlarged to~~ single-modal implementation uses $s_s = 1/2$, $s_i = 1/4$, $s_a = 1/8$, $\Sigma = \{1, 3, 5, 7, 9\}$ to ~~match~~ cover wider visible fractures, with the same radius $R = 3$ as in the preliminary study.

D. Graph Extraction

The extraction pipeline has four steps. First, ~~a thresholding removes low-confidence elements~~. ~~Pixels are filtered by positive curvature, then edges are filtered by the similarity S_{ij} . This pruning step is tuned with a single~~. A single retention factor τ parameter (the proportion of kept nodes and edges) is applied to both graph elements: the strongest fraction τ of edges is retained according to S_{ij} ; nodes are then ranked by their maximum incident similarity, and the strongest fraction τ is retained.

Second, connected components are obtained from the remaining graph. ~~The two following steps (third and fourth, below) are applied component-wise~~ On FIND, where the annotation targets one main crack, we retain the largest component. When several independent networks are of interest, the next two steps are applied to each retained component.

Third, a minimum spanning tree $\mathcal{T}$ [17] is computed using the dissimilarity d_{ij} . The tree preserves the “cheapest” coherent connections and turns the locally thick response into a structure suitable for centerline extraction.

Fourth, we compute a similarity-weighted tree centrality, ~~related to~~ betweenness centrality [18]. ~~Unlike general graphs~~

~~where betweenness centrality is computationally expensive~~ Unlike general-graph betweenness [19], ~~our reduction to a tree allows for a highly efficient~~ this score is computed in $O(|V|)$ linear-time computation ~~via~~ with a breadth-first search. ~~In our tree case, removing a vertex v splits $\mathcal{T}$ into q branches $\{b_1, \dots, b_q\}$. We define the similarity-weighted mass of branch b_r as~~

$$M(b_r) = \sum_{e \in b_r} S_e,$$

traversal. Each vertex receives the weight

$$w_i = \max_{j: (i, j) \in E} S_{ij}. \quad (13)$$

~~and compute the centrality by~~

$$C_B(v) = \sum_{1 \leq r < s \leq q} M(b_r) M(b_s).$$

~~Pixels close to the central axis connect large branch masses and therefore obtain high centrality. Peripheral pixels and short spurious branches have lower scores and are removed by a relative pruning threshold~~ After rooting the tree, let $\mathcal{T}_i$ be the subtree below x_i , $m_i = \sum_{x_k \in \mathcal{T}_i} w_k$, and $M = \sum_k w_k$. We use

$$C_{\mathcal{T}}(x_i) = m_i(M - m_i). \quad (14)$$

The highest-scoring vertex becomes the tree root, and the subtree masses are recomputed from it. The non-root scores follow (14) and are normalized, while the selected root is assigned score one. A branch is stopped when a child falls below the relative or absolute cutoff. Pixels on the main axis therefore remain, while short peripheral branches are removed.

Figure 2 ~~illustrates the complete chain~~ shows the intermediate maps for a FIND example. The positive-curvature map is close to a classical pixel-wise FRANGI response; ~~but~~ the graph similarity is cleaner because it also enforces pairwise orientation compatibility.

E. Role of the Components

The summary in Table I makes explicit which failure mode each term is intended to control.

TABLE I: Role of the components in the extraction chain.

Component	Role in the extraction chain
S_{shape}	Penalizes blob-like responses and favors elongated local structures.
S_{int}	Strong positive curvature; closest part to classical FRANGI evidence.
S_{align}	Penalizes locally incompatible directions and accidental bridges.
ρ_{ij}	Discourages long links between distant but visually similar pixels.
Weighted centrality	Selects pixels lying on the main centerline of the retained graph.

The ~~scale set sensitivity~~ parameters were selected empirically, starting from our preliminary study [8]. We retained s_i and s_a and increased s_s from $1/2$ to 2 for FIND and the Palais des Papes, allowing more variation in the FRANGI ratio, which is less stable than the curvature and orientation terms.

The scale set, radius and retention factor are fixed per dataset. We set Σ and radius from the expected crack-width range and R are the most physically interpretable parameters. Both should be chosen *ad-hoc* according to the expected crack width and the maximum gap to bridge.

The parameters s_s, s_i, s_a (which can be seen as standard deviation terms associated to $S_{\text{shape}}, S_{\text{int}}, S_{\text{align}}$, respectively) control the softness of the three compatibility terms. These are kept fixed across all experiments to demonstrate the transferability of our approach.

Finally, the threshold parameter from the largest gap that should be bridged; τ strictly governs the sparsity-recall controls the sparsity-recall trade-off of the extracted network. While the output topology naturally depends on this density constraint, we demonstrate that a fixed, conservative value ($\tau = 30\%$) yields robust extractions across all our diverse experimental setups. We use $\tau = 0.25$ for FIND and the Palais des Papes, and $\tau = 0.30$ for Vaches Noires, following the preliminary study [8].

IV. EXPERIMENTS

A. Datasets and Protocol

The quantitative evaluation uses FIND [6], a public benchmark with aligned intensity (I) and range (R) images for road-crack detection (and a third modality, ‘filtered’ F , which is a clean post-processing of the range modality). We compare against CrackSegDiff [12], the state-of-the-art supervised diffusion-based model trained a supervised diffusion model for multi-modal crack segmentation. For both compared methods, the same evaluation protocol is used: Dense masks are skeletonized before comparison and overlap metrics are computed with a fixed spatial tolerance.

We also report experiments on two geological image sets provided by Cerema. The first consists of two visible UAV images contains two annotated 512×512 visible UAV patches from the Vaches Noires landslide at Villers-sur-Mer (France), provided by a French national public institute for infrastructure (anonymized for review). The site contains badland morphology and multi-scale fracture networks. We call these evaluation patches Test1 and Test2; they are separate from the 3760×2058 image used to adapt the U-Net baseline. The second set is a photogrammetric intensity-range pair acquired on the retaining rock wall of the Palais des Papes in Avignon (France), also provided by the same institute.

B. Metrics

For a ground truth set A and a predicted set B , we use the JACCARD index [20], also known as Intersection over Union [21],

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}, \quad (15)$$

and the TVERSKY index [22],

$$T(A, B) = \frac{|A \cap B|}{|A \cap B| + \alpha |A \setminus B| + \beta |B \setminus A|}, \quad (16)$$

with $\alpha = 1$ and $\beta = 0.5$, so missed cracks are penalized more than over-detections. Since skeletons are one-pixel-wide, we dilate both skeletons by 3 pixels before computing these overlap scores. This tolerance is empirical and only compensates for small localization shifts at the 256×256 FIND resolution. We also report a discrete WASSERSTEIN distance [23], [24], which measures spatial displacement between predicted and reference skeleton pixels.

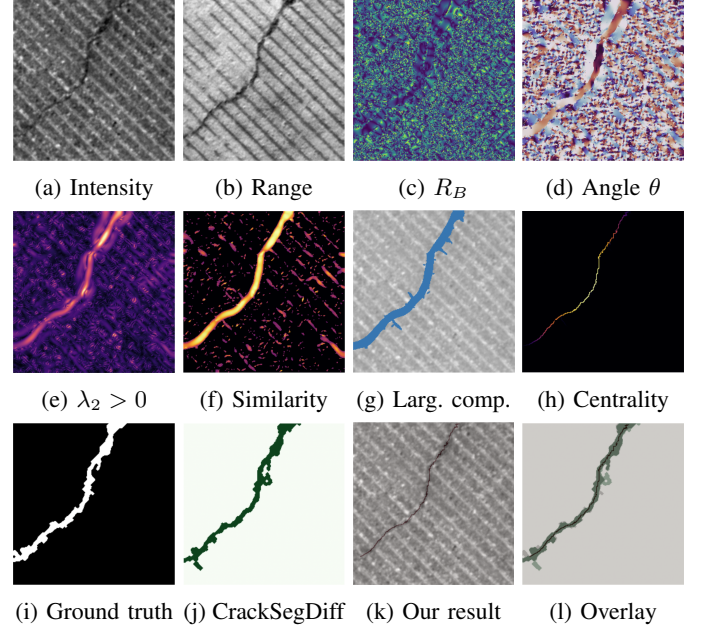


Fig. 2: Processing of the first FIND image example. **Top row:** (a) Top: Intensity and (b) Range modalities, (c) FRANGI ratio $R_B = |\lambda_1|/|\lambda_2|$ and (d) map of $\sin(\theta)$ associated to the Hessian orientation. **Middle row:** (e) Map $x \mapsto \lambda_2 \mathbf{1}_{\lambda_2 > 0}$ (very close to the original FRANGI filter) Middle: Positive curvature, (f) Our FRANGI similarity ($x_i \mapsto \max_{j, x_i \sim x_j} S_{ij}$), (g) the largest cluster component and (h) the weighted betweenness-centrality. **Bottom row:** (i) Bottom: Ground truth, (j) Result of CrackSegDiff on fused modalities, ours (k) Our resulting skeleton (JACCARD / IoU: 63% ; TVERSKY: JACCARD, 71% ; WASSERSTEIN: TVERSKY, 8px WASSERSTEIN) and (l) Overlay of ground truth, CrackSegDiff and our skeleton overlay.

C. FIND: Clean and Noisy Data

Table II compares the proposed method with CrackSegDiff on clean FIND images using intensity and range. As expected, CrackSegDiff achieves the highest overlap on the clean, in-distribution data it was trained on. However, our geometric formulation still recovers the main topology of the fracture networks without requiring any training data, demonstrating its viability as a training-free baseline.

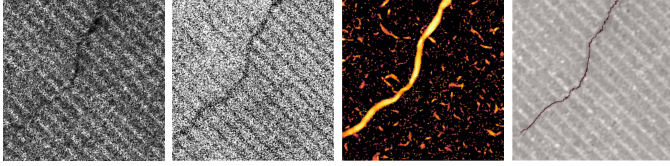
The FIND dataset [6] also contains a filtered range modality. Table III reports a modality contribution analysis for the proposed method. The filtered modality alone is strong (emphasizing the need, which shows the importance of

TABLE II: Performance on clean FIND data

Method	JACCARD	TVERSKY	WASSERSTEIN
Ours ($I + R$)	60%	68%	18 px
CrackSegDiff	87%	90%	5 px

TABLE III: Modality contribution analysis for the proposed method on FIND. I : Intensity, R : Range, F : Filtered range.

Weights	JACCARD $\uparrow$	TVERSKY $\uparrow$	WASSERSTEIN $\downarrow$
$I = 1$	41%	48%	43 px
$R = 1$	54%	62%	20 px
$F = 1$	60%	68%	12 px
$I = \frac{1}{2}, R = \frac{1}{2}$	60%	68%	18 px
$I = \frac{1}{3}, R = \frac{1}{3}, F = \frac{1}{3}$	63%	71%	11 px



(a) Noisy Int. (b) Noisy Range (c) Similarity (d) Our result

Fig. 3: Noisy FIND example ($s = 0.5$) with speckle noise on intensity and Gaussian noise on range. The graph similarity remains concentrated around the main crack.

clean, possibly pre-processed ~~modalities with our method~~, and fusing inputs for our method. Fusing intensity, range and filtered range gives the best result. This and supports the operator-level fusion strategy.

To evaluate the framework’s robustness ~~outside the clean training distribution~~ to synthetic sensor noise, we artificially degrade the input modalities ~~using sensor-specific noise models~~. Specifically, we perturb the intensity modality with multiplicative speckle noise to simulate surface reflectance variations, and corrupt the range modality with additive Gaussian noise to model instrumental measurement errors:

$$I_{\text{noisy}}(x) = I(x) \cdot (1 + N(x)), \quad (17)$$

$$R_{\text{noisy}}(x) = R(x) + N'(x), \quad (18)$$

where $I(x)$ and $R(x)$ denote the original intensity and range signals at pixel x , respectively. $N(x), N'(x) \sim \mathcal{N}(0, s^2)$ are independent Gaussian random variables where s^2 measures the variance of the noise. This protocol evaluates the robustness of our operator-level fusion ~~against domain shifts and severe sensor noise~~ to controlled sensor perturbations.

Figure 3 shows an example at a strong noise level; Figure 4 reports the degradation curves. CrackSegDiff initially performs better, but its score decreases rapidly as noise increases. The proposed method starts lower but degrades more smoothly because the graph uses multi-scale curvature and orientation compatibility instead of learned appearance statistics alone.

D. Geological Domain Shift

TABLE IV: Quantitative comparison on the two annotated Vaches Noires patches.

Image	Method	JACCARD $\uparrow$	TVERSKY $\uparrow$	WASSERSTEIN $\downarrow$
Test1	Ours	0.47	0.52	34 px
	U-Net + T.L.	0.50	0.60	38 px
Test2	Ours	0.29	0.32	81 px
	U-Net + T.L.	0.34	0.36	60 px

~~Our Vaches Noires dataset provides only visible data for a geological case, different~~ The Vaches Noires data differ markedly from the road-crack FIND images. For these images, we use a single visible modality and several scales $\Sigma = \{1, 3, 5, 7, 9\}$. The other extraction parameters are set *ad hoc*¹: $R = 3$, and contain only a visible modality. We use $\Sigma = \{1, 3, 5, 7, 9\}$, $R = 3$ and $\tau = 0.3$ for both evaluation patches.

We compare against a U-Net baseline adapted by transfer learning [25] on an annotated 3760×2058 pixel image from the same area, following the common strategy used when only a small amount of domain-specific annotation is available. This ground truth annotation required approximately two days of work by ~~a geophysicist expert~~ an expert geophysicist. The proposed method uses only the visible image and no training annotation.

~~Table IV reports the quantitative overlap metrics.~~ The supervised baseline gives slightly higher overlap scores on both ~~images~~ patches, which is expected ~~since the neural network because it was adapted on a very similar image (via transfer learning), annotated in the exact same context.~~ By contrast, the proposed graph ~~nearby~~ annotated image. Our method uses no annotation and ~~more directly isolates the main network backbone.~~ On Test1, this even yields gives a lower WASSERSTEIN distance. ~~The qualitative comparisons shown in Figures 5 and 6 visually confirm this.~~ These results demonstrate that our geometric formulation offers a competitive baseline for domain shifts, bypassing the ~~costly expert annotations required to fine-tune deep learning model~~ on Test1. These two patches are useful transfer tests, but they are not enough to establish performance across domains.

~~For a second qualitative transfer case, we apply the same~~ We also apply the method to a fractured limestone wall at the Palais des Papes, using an orthoimage and a photogrammetric depth map. ~~The wall contains a dominant vertical fracture and smaller intersecting cracks.~~ Figure 7 shows that the main fracture ~~network~~ is recovered without retraining. ~~Minor~~; minor cracks are missed, mostly where depth resolution and contrast are insufficient.

CODE AND DATASET DATA AVAILABILITY

~~To maintain anonymity during the double-blind review process, the source code and datasets are withheld from~~

¹As discussed in Section III-E, the most important parameter to tune is the scale set Σ , which can be easily adjusted *ad hoc* using only prior knowledge regarding the expected width range of the structures of interest, i.e., the cracks.

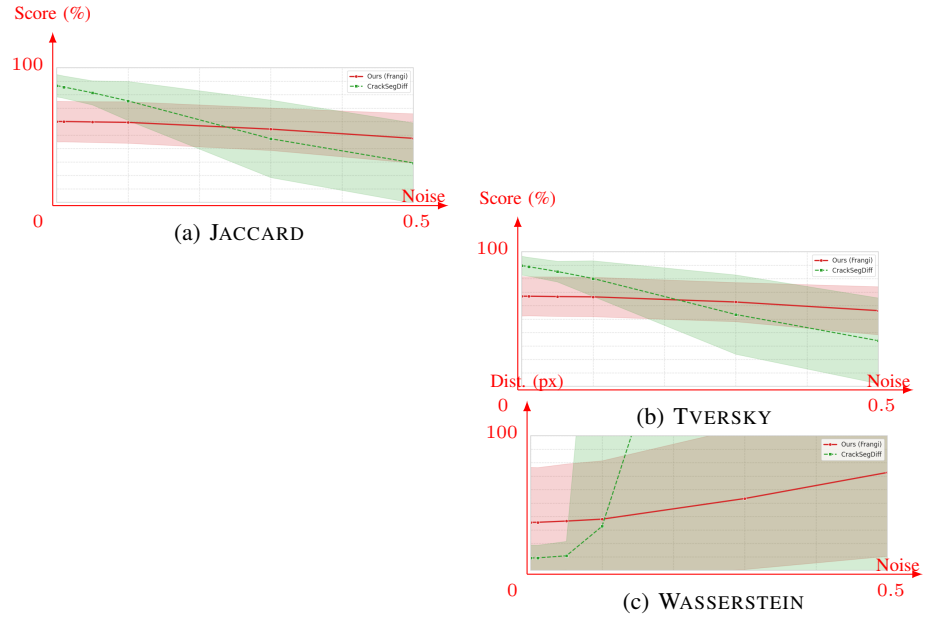


Fig. 4: Performance degradation on noisy FIND. Red: Proposed method; green: CrackSegDiff. Shaded regions show standard deviation. The horizontal axis is the noise level s (standard deviation of the Gaussians).

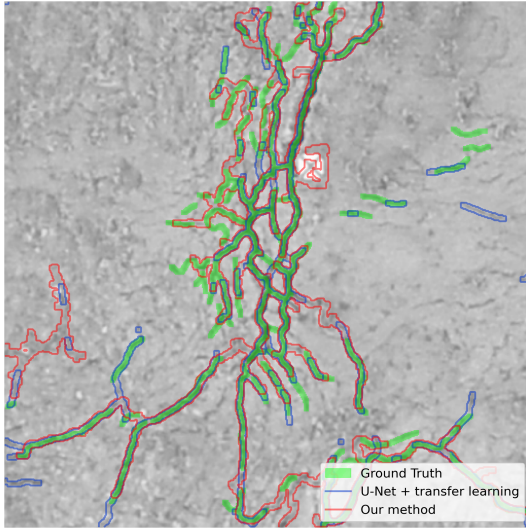


Fig. 5: ~~Qualitative comparison on the Test1 Vaches Noires image (France)Test1. The ground-Truth is shown in~~ : green, ~~our method in~~ ; ours: red, ~~and~~ ; U-Net + transfer learning ~~in~~ : blue.

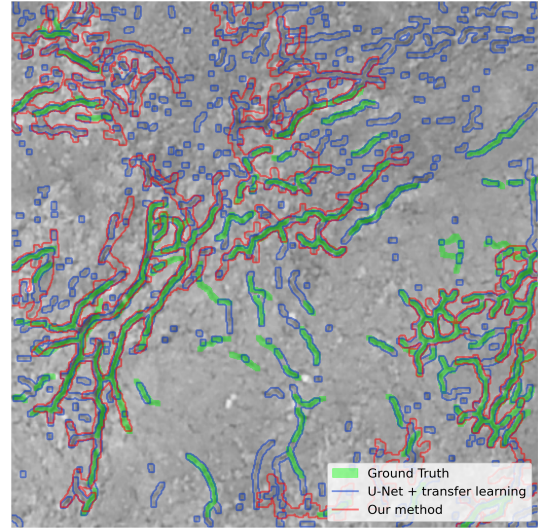


Fig. 6: ~~Qualitative comparison on the Test2 Vaches Noires image (France)Test2. The ground-Truth is shown in~~ : green, ~~our method in~~ ; ours: red, ~~and~~ ; U-Net + transfer learning ~~in~~ : blue.

~~this submission. They will be made publicly available upon~~
~~acceptance. The code and reproduction notebooks are available~~
~~at~~ <https://github.com/Ayana-Inria/Frangi-EUVIP>. The FIND
benchmark is publicly available [6]. The non-public data used
in our experiments are available at [https://drive.google.com/](https://drive.google.com/drive/folders/1iC7QrdB2vZmaunjjPrIcu9r5wMeBPMlk)
[drive/folders/1iC7QrdB2vZmaunjjPrIcu9r5wMeBPMlk](https://drive.google.com/drive/folders/1iC7QrdB2vZmaunjjPrIcu9r5wMeBPMlk).

V. CONCLUSION

We presented a ~~purely~~ geometric, training-free ~~framework~~
method for multi-modal crack skeletonization. ~~By lifting the~~
classical FRANGI filter to a pairwise spatial graph, our method

~~enforces structural coherence to extract a reliable centerline.~~
While deep learning models dominate on their clean training
domains, our approach provides a robust, annotation-free
alternative that generalizes well to entirely new domains
(e.g., from urban roads to geological fault networks) On clean
FIND data, the supervised model performs better; under added
noise, our graph method degrades more slowly. The geological
results show that the method can be applied without retraining,
but the evaluation is limited to two annotated patches and one
qualitative case.

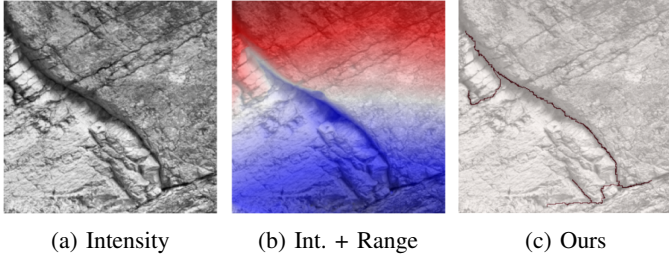


Fig. 7: ~~Retaining rock wall of the Palais des Papes (France).~~
~~Left: Intensity image. Middle: Intensity with intensity,~~
~~depth overlay. Right: Extracted and extracted skeleton.~~

~~Furthermore, because our framework explicitly outputs a~~
~~The output is an editable spatial graph, the extracted skeleton~~
~~can be easily edited by domain experts. This makes it a~~
~~practical tool for the~~ which is useful for semi-automatic
~~annotation of datasets intended for supervised models.~~

~~Extending this graph formulation to capture higher-order~~
~~topological interactions is a promising direction for future~~
~~work and make our approach even more robust. We also plan~~
~~to benchmark our approach against adapted vision models.~~
~~Vision foundation models are now essential baselines for crack~~
~~segmentation; a direct comparison with SAM [26] and its~~
~~crack-specific adaptation CrackSAM [27] (see also by the~~
~~team of CrackSegDiff) and conduct a sensitivity analysis on~~
~~other is therefore a priority. We will also test sensitivity on~~
~~established datasets [28] and higher-order graph variants [29].~~

ACKNOWLEDGMENT

The work of the Inria authors was partially supported by
 Bpifrance through the LiChIE contract (2020–2025). The first
 author was supported by Université Côte d’Azur through
 DS4H (ANR-17-EURE-0004) and 3IA (ANR-19-P3IA-0002).

REFERENCES

- [1] E. Bird, *Coastal cliffs: morphology and management*. Springer, 2016.
- [2] B. Jafrasteh, I. Manighetti, and J. Zerubia, "Generative adversarial networks as a novel approach for tectonic fault and fracture extraction in high resolution satellite and airborne optical images," in *XXIV International Society of Photogrammetry and Remote Sensing Congress*, vol. XLIII-B3-2020, Nice, France, Aug. 2020, pp. 1219–1227.
- [3] L. Mattéo, I. Manighetti, Y. Tarabalka, J.-m. Gaucel, M. van den Ende, A. Mercier, O. Tassar, N. Girard, F. Leclerc, T. Giampetro, S. Dominguez, and J. Malavieille, "Automatic fault mapping in remote optical images and topographic data with deep learning," *Journal of Geophysical Research : Solid Earth*, Apr. 2021.
- [4] X. Zhang, H. Wang, Y.-A. Hsieh, Z. Yang, A. Yezzi, and Y.-C. Tsai, "Deep learning for crack detection: A review of learning paradigms, generalizability, and datasets," 2025, preprint arXiv:2508.10256.
- [5] A. F. Frangi, W. J. Niessen, K. L. Vincken, and M. A. Viergever, "Multiscale vessel enhancement filtering," in *Medical Image Computing and Computer-Assisted Intervention*. Springer, 1998.
- [6] S. Zhou, C. Canchila, and W. Song, "Fused Image dataset for convolutional neural Network-based crack Detection (FIND)," Mar. 2022, <https://zenodo.org/records/6383044>.
- [7] Y. Sato, S. Nakajima, N. Shiraga, H. Atsumi, S. Yoshida, T. Koller, G. Gerig, and R. Kikinis, "Three-dimensional multi-scale line filter for segmentation and visualization of curvilinear structures in medical images," *Medical Image Analysis*, vol. 2, no. 2, pp. 143–168, 1998.
- [8] L. Hauseux, R. Antoine, P. Foucher, P. Charbonnier, and J. Zerubia, "Généralisation du filtre de Frangi pour l'extraction des réseaux de fissures au sein d'images d'un glissement de terrain. Comparaison avec une méthode d'apprentissage profond," in *XXe Colloque Francophone de Traitement du Signal et des Images (GRETSI)*, Strasbourg, Aug. 2025, pp. 389–392, Online.
- [9] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," *MICCAI*, 2015.
- [10] Q. Zou, Z. Zhang, Q. Li, X. Qi, Q. Wang, and S. Wang, "DeepCrack: Learning hierarchical convolutional features for crack detection," *IEEE Transactions on Image Processing*, vol. 28, no. 3, pp. 1498–1512, 2019.
- [11] H. Liu, X. Miao, C. Mertz, C. Xu, and H. Kong, "CrackFormer: Transformer Network for Fine-Grained Crack Detection," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 10 2021, pp. 3783–3792.
- [12] X. Jiang, L. Jiang, A. Wang, K. Zhu, and Y. Gao, "CrackSegDiff: Diffusion Probability Model-based Multi-modal Crack Segmentation," 2024, preprint arXiv:2410.08100.
- [13] T.-C. Lee, R. L. Kashyap, and C.-N. Chu, "Building skeleton models via 3-D medial surface/axis thinning algorithms," *Computer Vision, Graphics, and Image Processing: Graphical Models and Image Processing*, vol. 56, no. 6, pp. 462–478, 1994.
- [14] S. Van der Walt, J. L. Schönberger, J. Nunez-Iglesias, F. Boulogne, J. D. Warner, N. Yager, E. Gouillart, and T. Yu, "Scikit-image: image processing in Python," *PeerJ*, vol. 2, p. e453, 2014, Online.
- [15] T. Lindeberg, "Edge detection and ridge detection with automatic scale selection," *International Journal of Computer Vision*, vol. 30, no. 2, pp. 117–156, 1998.
- [16] R. A. Horn and C. R. Johnson, *Matrix analysis*, 2nd ed. New York, NY, USA: Cambridge University Press, 2012.
- [17] R. C. Prim, "Shortest connection networks and some generalizations," *The Bell System Technical Journal*, vol. 36, no. 6, pp. 1389–1401, 1957.
- [18] L. C. Freeman, "A set of measures of centrality based on betweenness," *Sociometry*, pp. 35–41, 1977.
- [19] U. Brandes, "A faster algorithm for betweenness centrality," *Journal of Mathematical Sociology*, vol. 25, no. 2, pp. 163–177, 2001.
- [20] P. Jaccard, "Distribution de la flore alpine dans le bassin des Dranses et dans quelques régions voisines," *Bulletin De La Société Vaudoise Des Sciences Naturelles*, 1901.
- [21] M. Berman, A. R. Triki, and M. B. Blaschko, "The Lovász-Softmax loss: A tractable surrogate for the optimization of the intersection-over-union measure in neural networks," *CVPR*, 2018.
- [22] A. Tversky, "Features of similarity," *Psychological Review*, vol. 84, no. 4, p. 327–352, 1977.
- [23] L. N. Vaserstein, "Markov processes over denumerable products of spaces, describing large systems of automata," *Problemy Peredači Informacii*, vol. 5, no. 3, pp. 64–72, 1969.
- [24] R. Flamary, C. Vincent-Cuaz, N. Courty, A. Gramfort, I. Redko *et al.*, "Pot: Python optimal transport library," <https://github.com/PythonOT/POT>, 2026.
- [25] Y. Shi, M. Ballesio, K. Johansen, D. Trentman, Y. Huang, M. F. McCabe, R. Bruhn, and G. Schuster, "Semi-universal geo-crack detection by machine learning," *Frontiers in Earth Science*, vol. 11, p. 1073211, 2023.
- [26] A. Kirillov, E. Mintun, N. Ravi, H. Mao, C. Rolland, L. Gustafson, T. Xiao, S. Whitehead, A. C. Berg, W.-Y. Lo *et al.*, "Segment anything," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023, pp. 4015–4026.
- [27] K. Ge, C. Wang, Y. Guo, Y. Tang, Z. Hu, and H. Chen, "Fine-tuning vision foundation model for crack segmentation in civil infrastructures," *Construction and Building Materials*, vol. 431, p. 136573, 2024.
- [28] Y. Shi, L. Cui, Z. Qi, F. Meng, and Z. Chen, "Automatic road crack detection using random structured forests," *IEEE Transactions on Intelligent Transportation Systems*, vol. 17, no. 12, pp. 3434–3445, 2016.
- [29] L. Hauseux, K. Avrachenkov, and J. Zerubia, "Generalization of single-linkage with higher-order interactions," *Applied Network Science*, vol. 11, no. 1, p. 9, 2026.